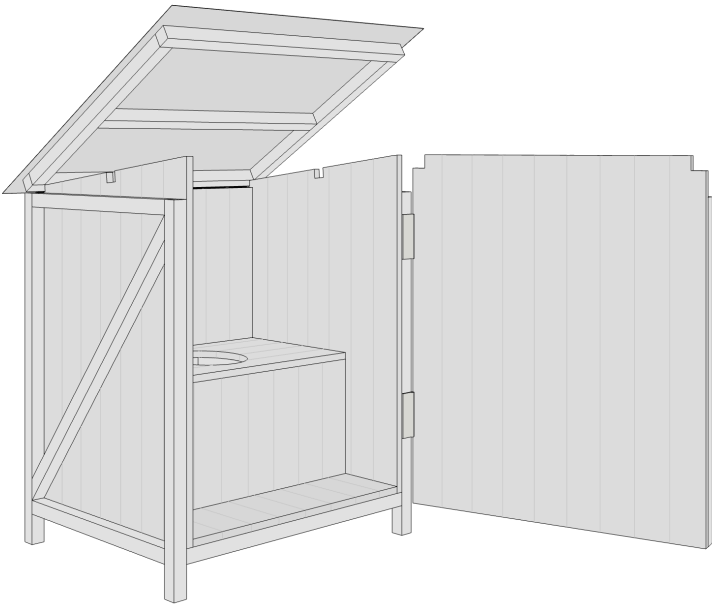


WORKING DRAWING

CAN AI BUILD A TOILET YET?

An experiment in re-drawing a technical drawing as an editable parametric CAD model with Claude and Codex.

PROJECT	DASS
ISSUE	2026-07-30
ARCHITECT	<u>@HANNES.SODERQUIST</u>
CODE	<u>@FEELEPXYZ</u>
SOURCE	 <u>GITHUB</u>
UNITS	MILLIMETRES
KERF	2.8 MM PER CUT
SHEETS	A-200 TO A-400



The finished building.

STRUCTURAL TIMBER

MATERIAL

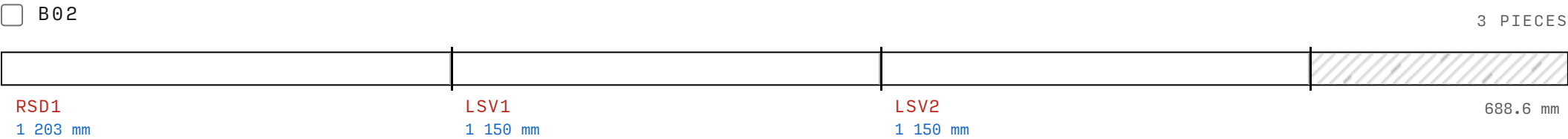
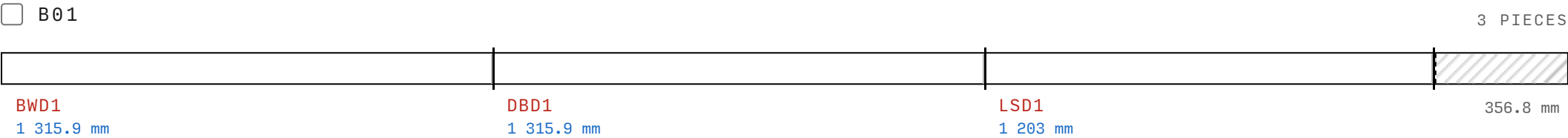
Frame timber, 45 × 45 × 4 200 mm. Quantity: 9 lengths.

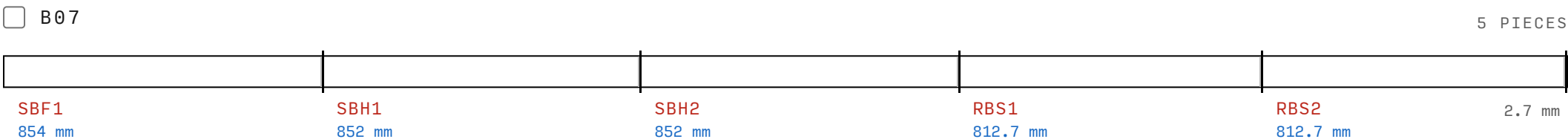
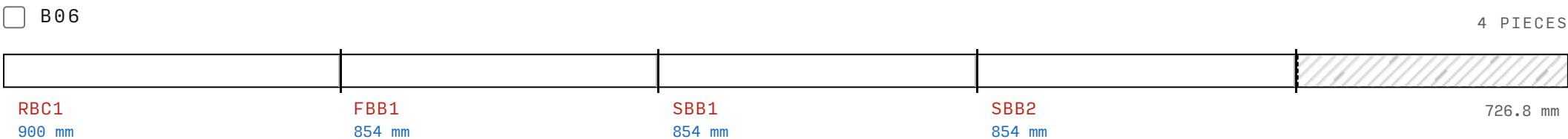
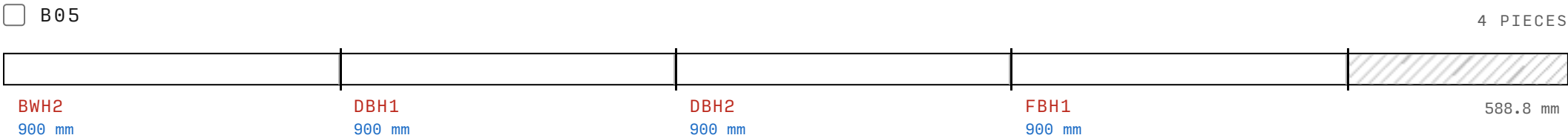
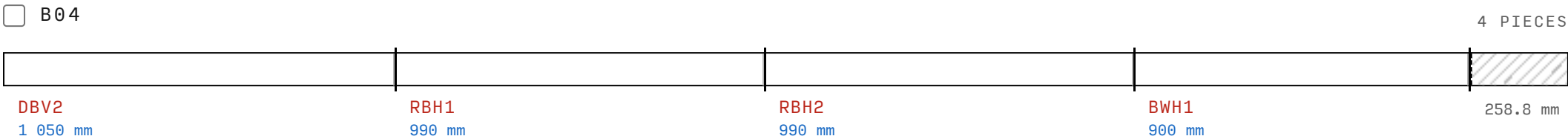
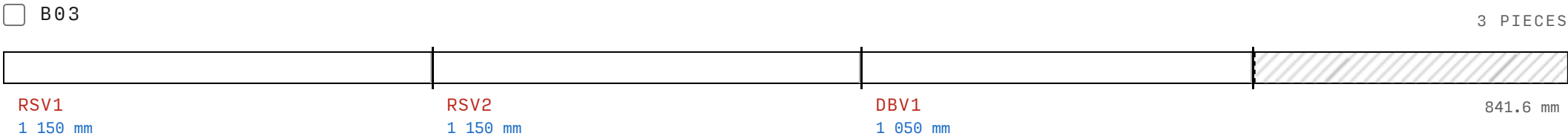
The 9 beam stock lengths below are named B01 to B09. Each cut diagram shows the marked beam pieces cut from that length.

Cut every piece at one stop setting before you change the stop. A batch that reaches the end of a stock length continues on the next length.

These sheets are exact only for this material size. One 2.8 mm kerf is used for every piece.

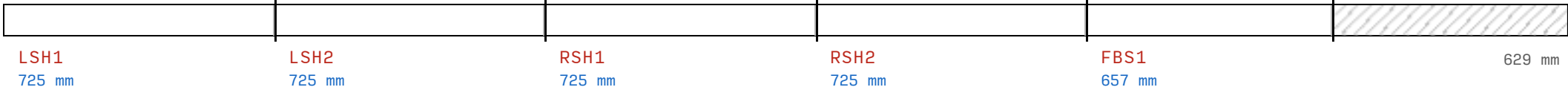
If your material ends are rough, shave a small amount off from each piece first.





☐ B08

5 PIECES



☐ B09

5 PIECES



RÅSPONT (MATCHBOARD/V-GROOVE CLADDING)

MATERIAL

Råspont (matchboard/V-groove cladding), 120 × 23 × 4 500 mm.
Quantity: 12 lengths.

The 12 cladding stock lengths below are named P01 to P12. Each cut diagram shows the marked boards cut from that length.

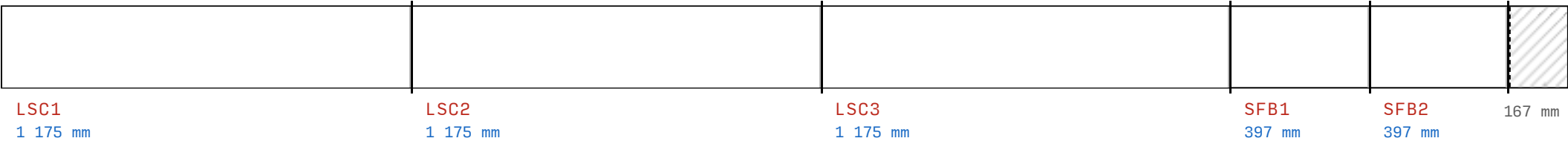
The first 8 stock lengths release all twenty-three 1 175 blanks.
Label the P01 to P03 remainders. Keep them for the final 397 pass.
Fourteen side boards come next, then nine door boards, at one stop setting.

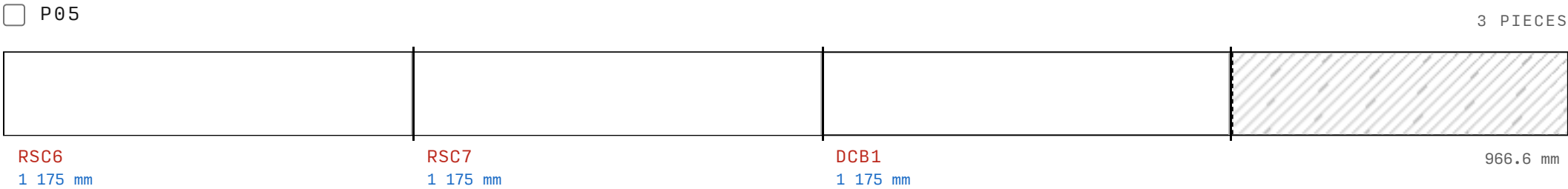
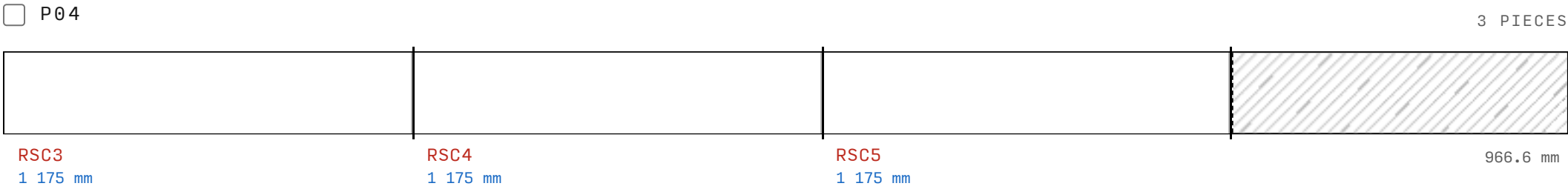
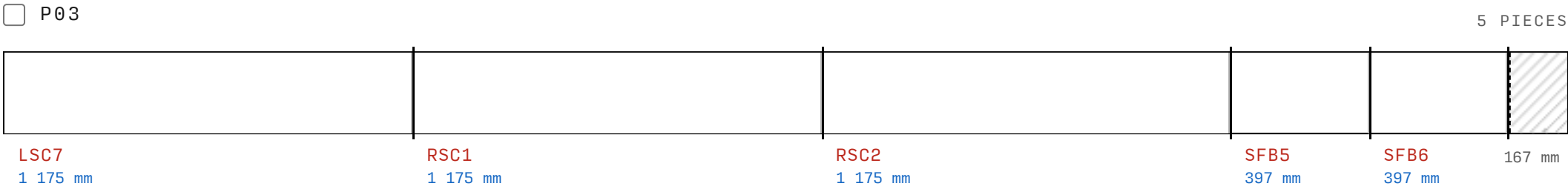
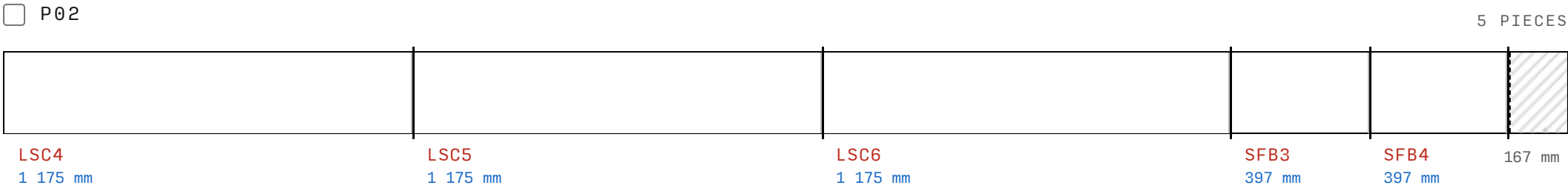
These sheets are exact only for this material size. One 2.8 mm kerf is used for every piece.

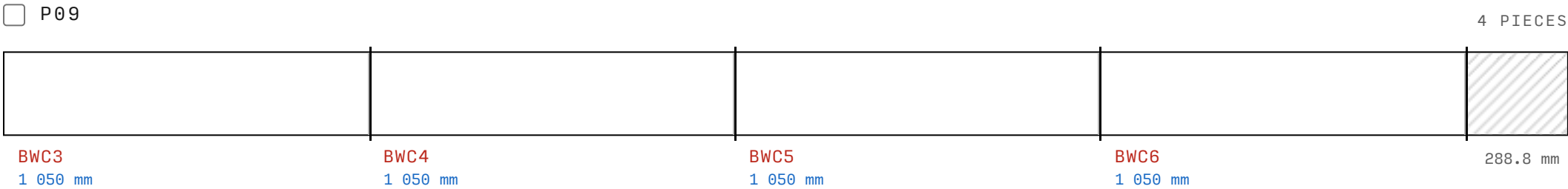
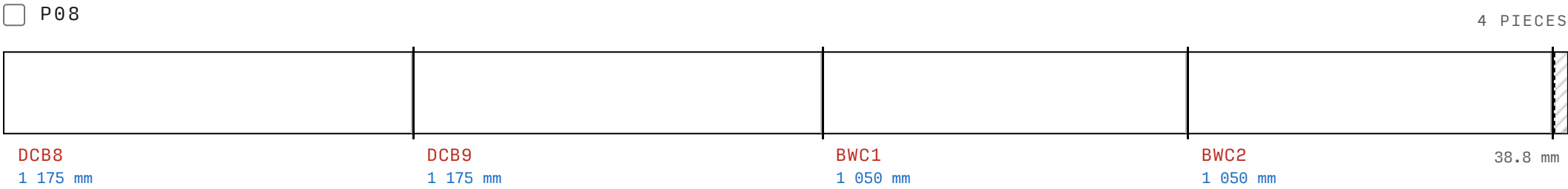
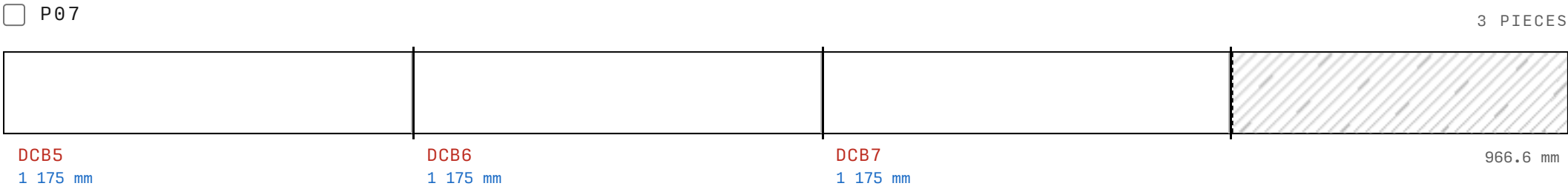
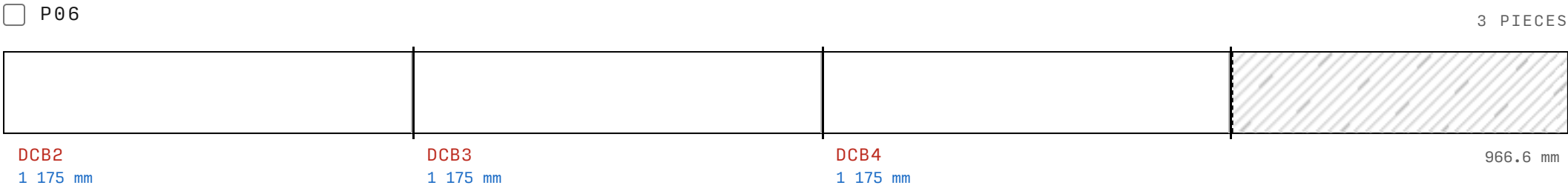
If your material ends are rough, shave a small amount off from each piece first.

☐ P01

5 PIECES

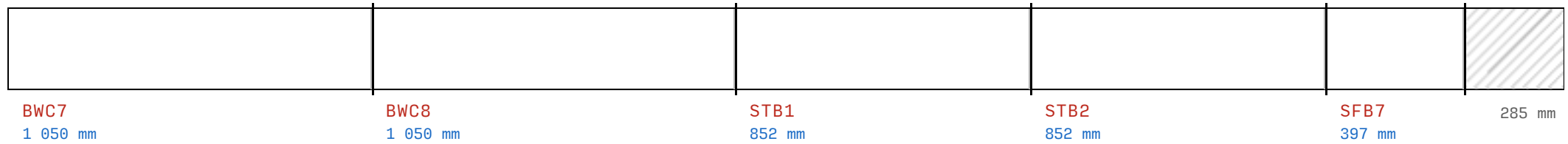






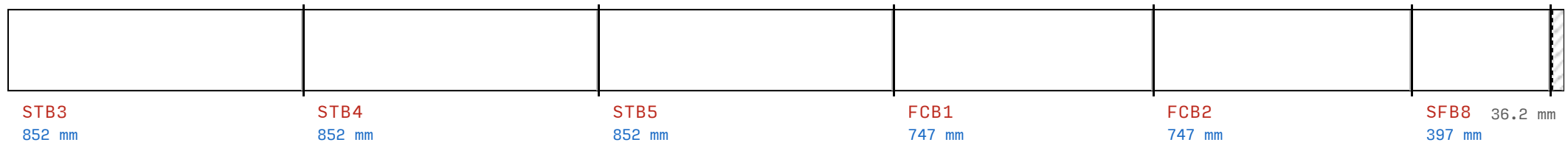
☐ P10

5 PIECES



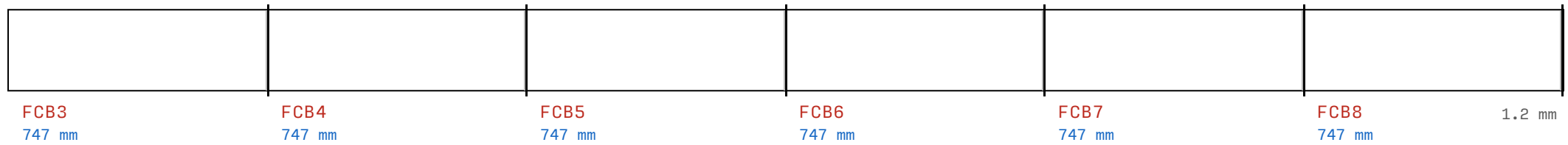
☐ P11

6 PIECES



☐ P12

6 PIECES



Gaps between boards change the joined width. Fix each board to its unit frame before you mark any final edge. Use the frame edge to set the circular-saw guide. Cut only after the boards are fixed. Sheet A-400 shows each estimated overhang and cut line.

UNIT DRAWINGS AND ASSEMBLY

Each unit opens with its general arrangement, dimensioned and coded off the model, then builds through its own numbered steps. A step draws one operation: work already done ghosts back, work this step touches goes solid, and the band under the drawing names the operation before the caption is read.

Black screw paths are measured 120 mm screws: each starts on the source beam and finishes in the centre of the receiving section. Diagonal paths follow the diagonal beam from its mitred end.

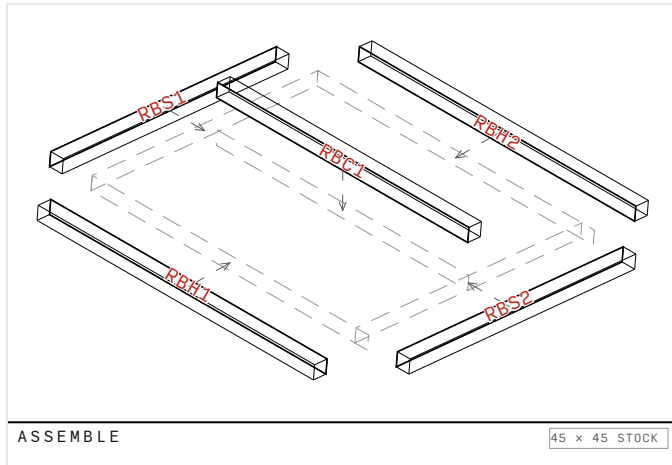
ASSEMBLY HARDWARE

Use these fasteners for the timber assembly.

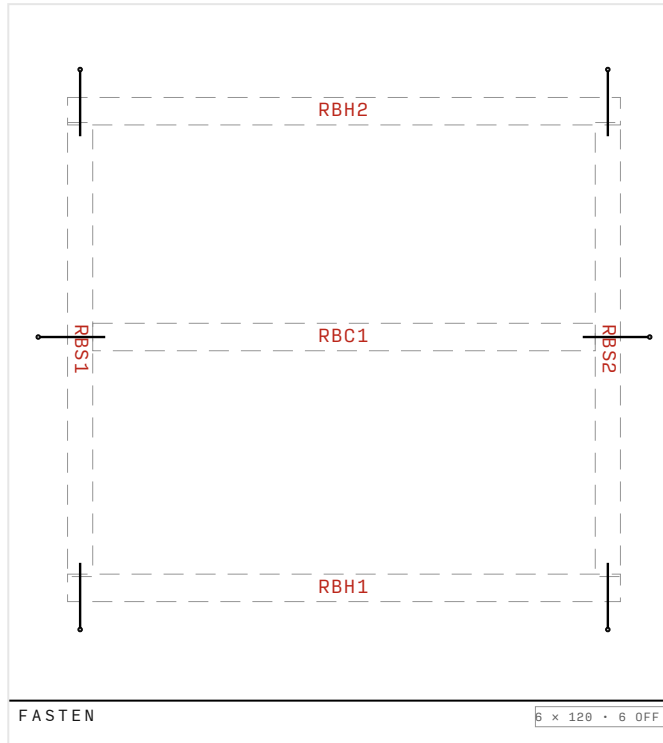
USE	FASTENER
Frame beams and braces	6 × 120 mm sunk wood screws
Beam-to-beam support joints, for example floor supports	6 × 90 mm sunk wood screws
Råspont (matchboard/V-groove cladding) to beams	2.8 × 60 mm nails or 6 × 60 mm sunk wood screws

HOLD

Fit the moving hinge leaf now. Hang the roof only after the shell is square.

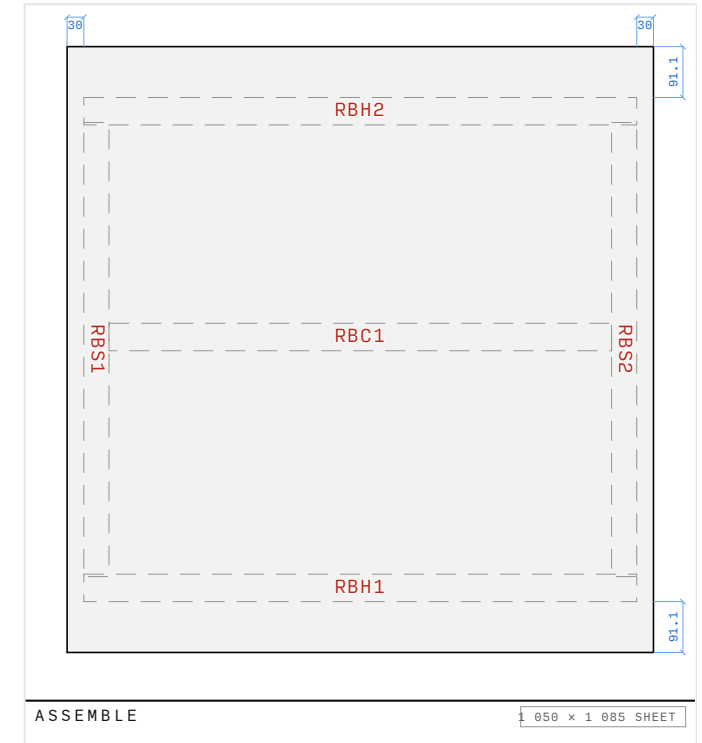
**1 LAY THE FIVE ROOF BEAMS**

• A-401.1

☐ DONE**2 SCREW THE FRAME CORNERS**

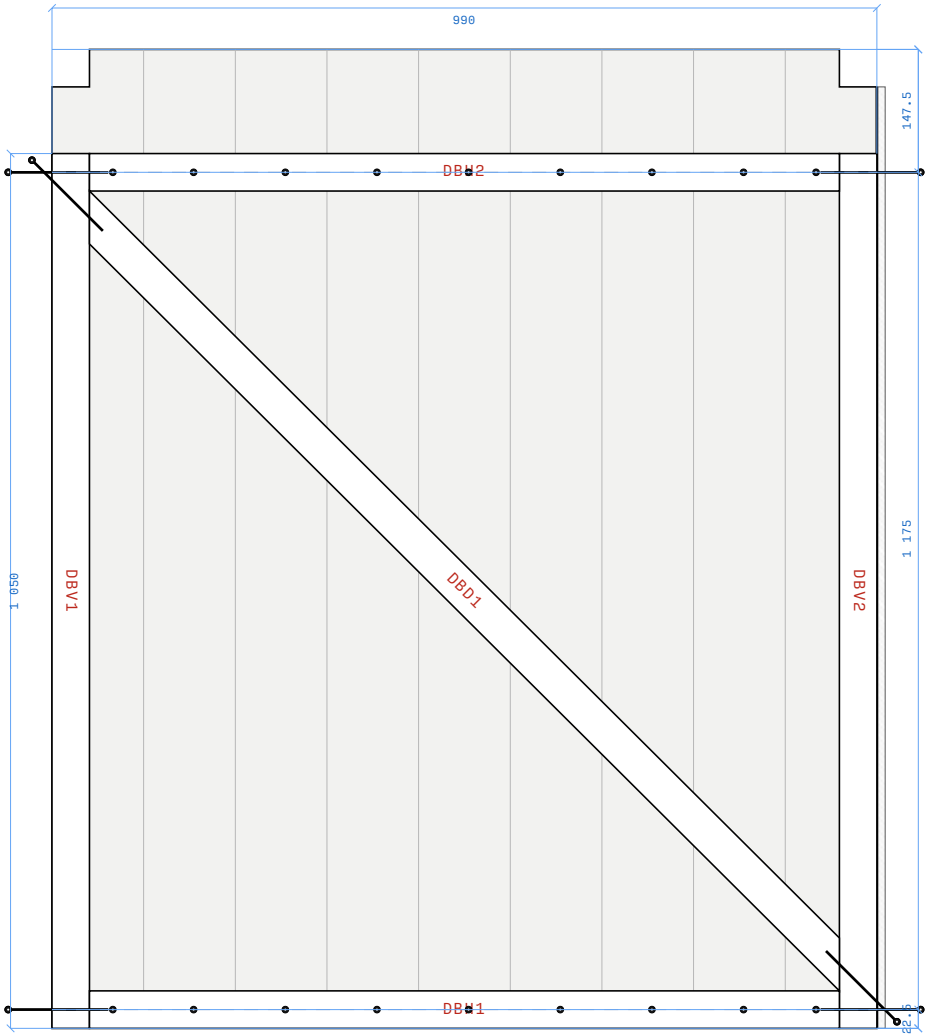
• A-401.2

Match the frame diagonals before you drive a screw.

☐ DONE**3 CENTRE THE METAL SHEET**

• A-401.3

☐ DONE

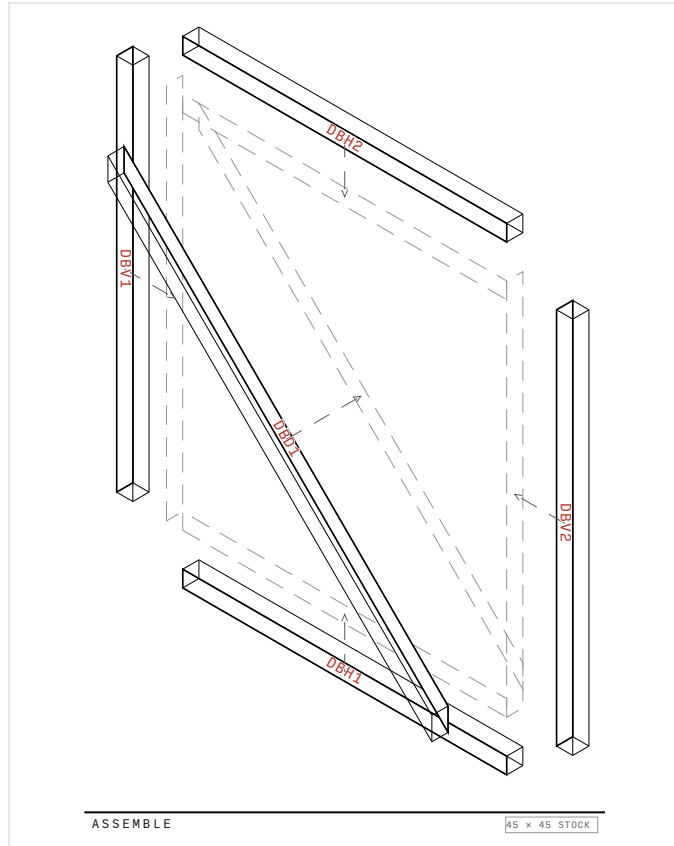


DOOR UNIT • A-402 • FRONT ELEVATION

seen from outside the door • hatched edge is the estimated
on-frame trim

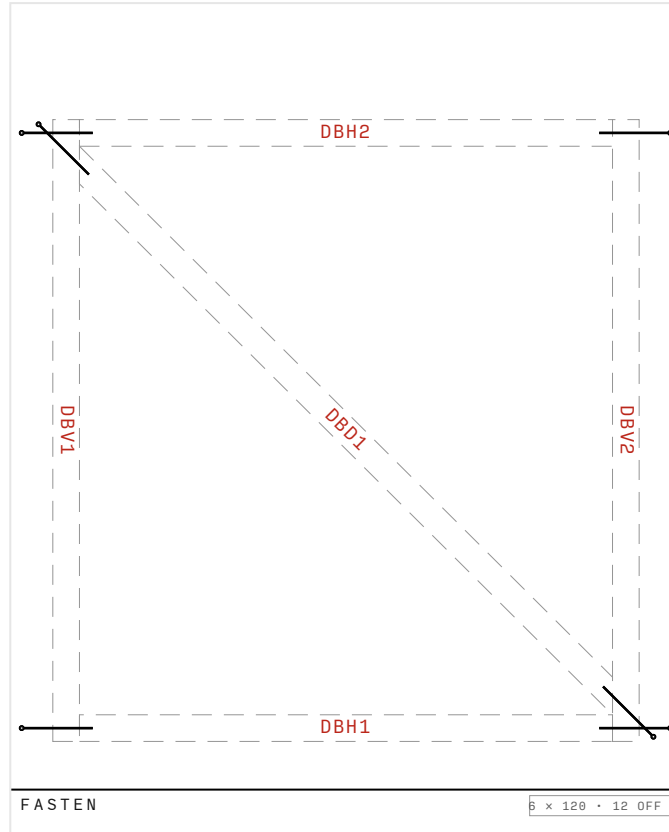
HOLD

Fit the moving hinge leaves now. Hang the door after the shell is square.



- 1 BUILD THE FRAME AND FIT DBD1
 - A-402.1

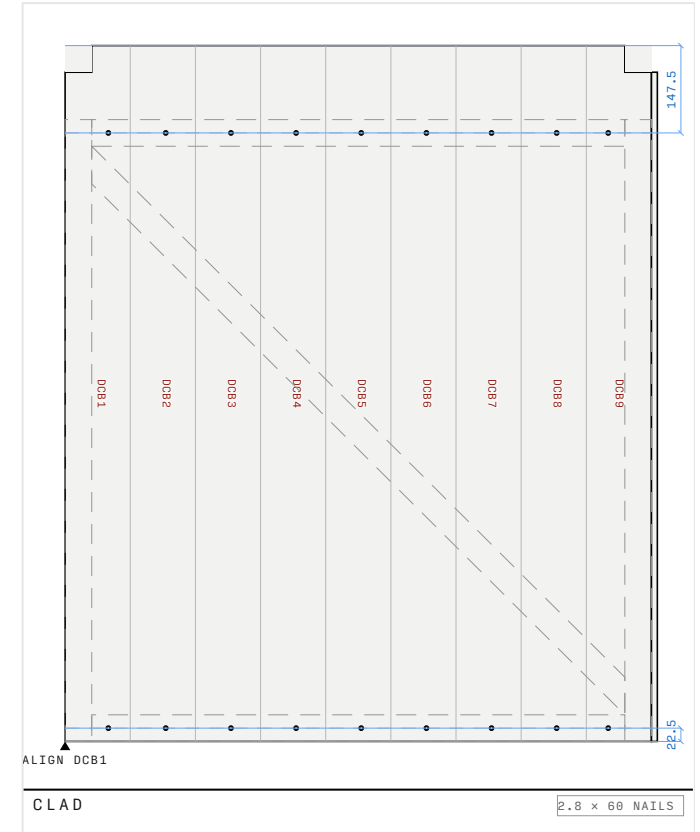
☐ DONE



- 2 SCREW THE DOOR FRAME
 - A-402.2

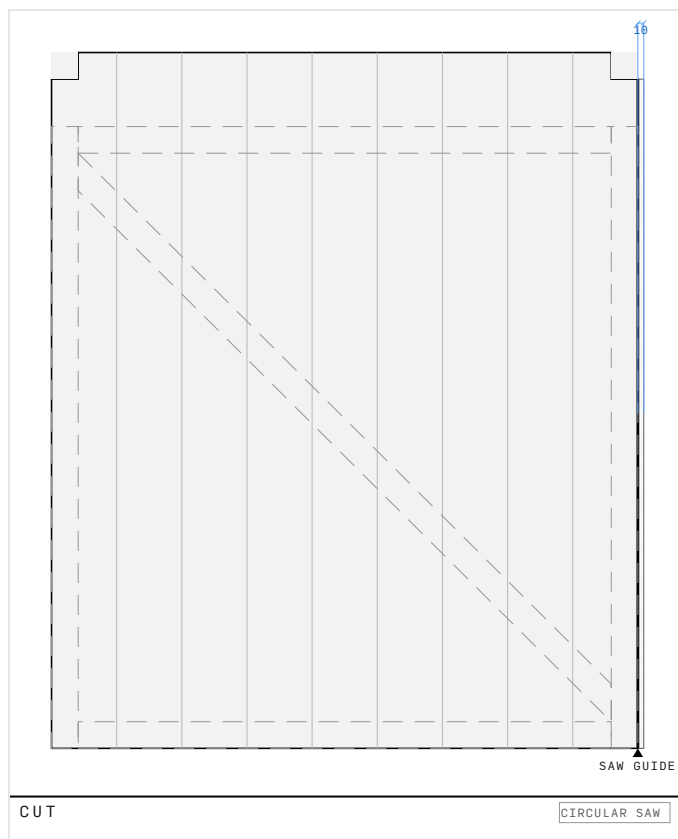
Match the frame diagonals before you drive a screw.

☐ DONE



- 3 FIX DCB1 TO DCB9 ON THE INSIDE FACE
 - A-402.3

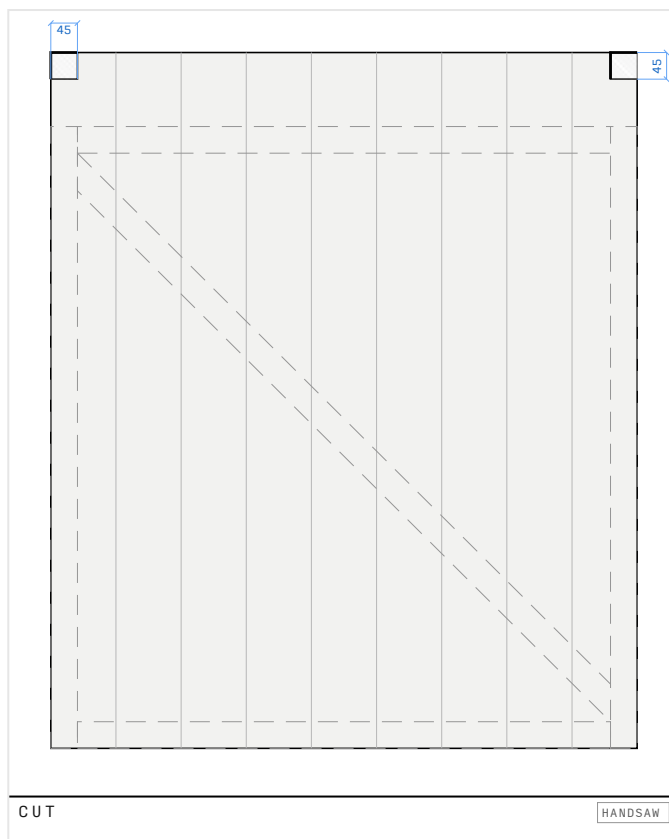
☐ DONE



4 TRIM THE RIGHT OVERHANG

• A-402.4

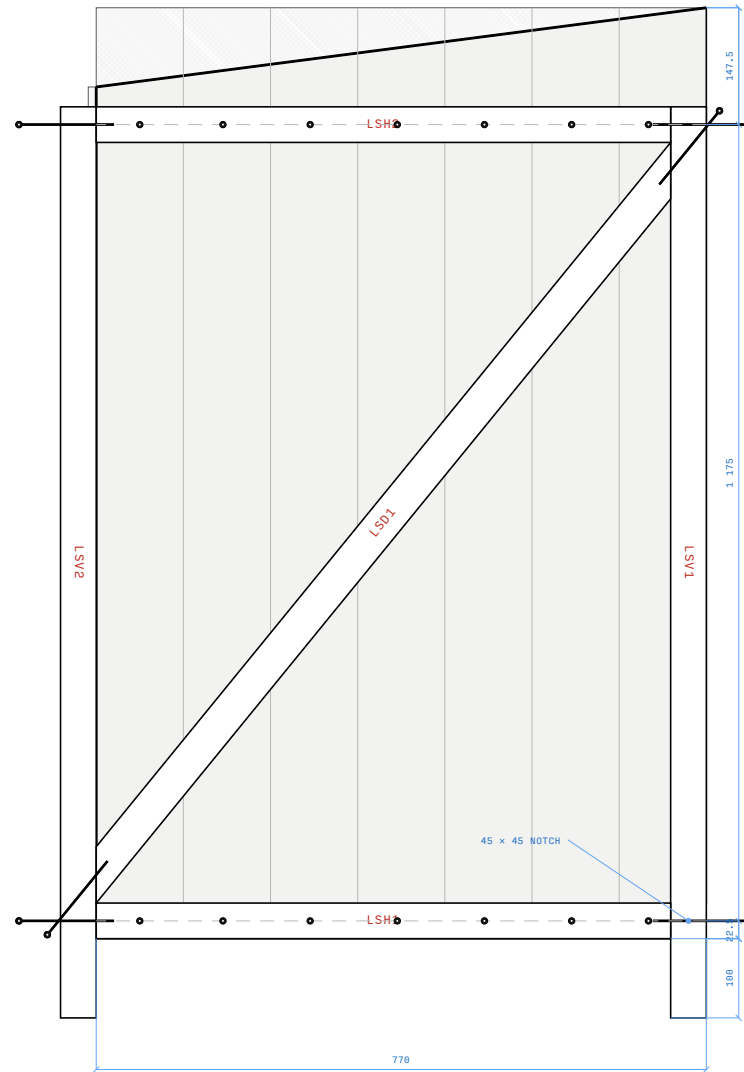
☐ DONE



5 CUT THE TWO TOP RELIEFS

• A-402.5

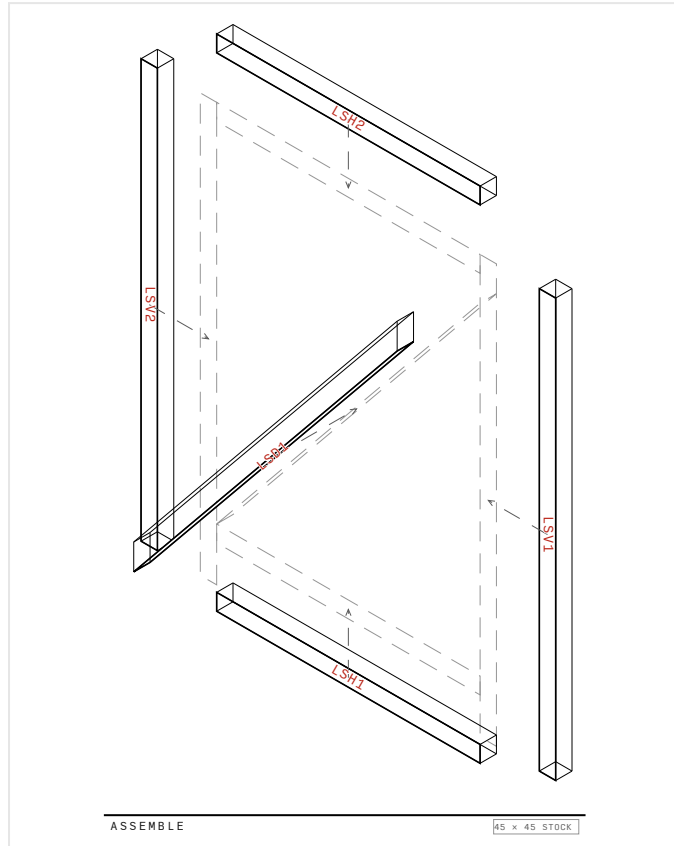
☐ DONE



LEFT SIDE UNIT • A-403 • LEFT ELEVATION
front edge to the right • 100 fall over 770 • 45 x 45
bottom-front notch

HOLD

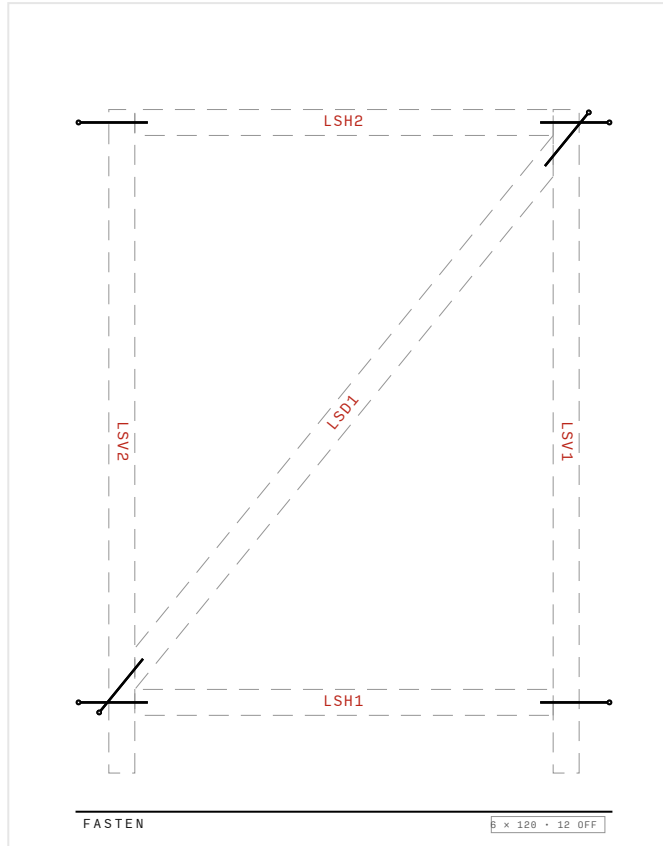
Do not pre-cut the roof reliefs. Hang and close the roof, then cut to the scribe.



1 BUILD THE SIDE FRAME

• A-403.1

☐ DONE

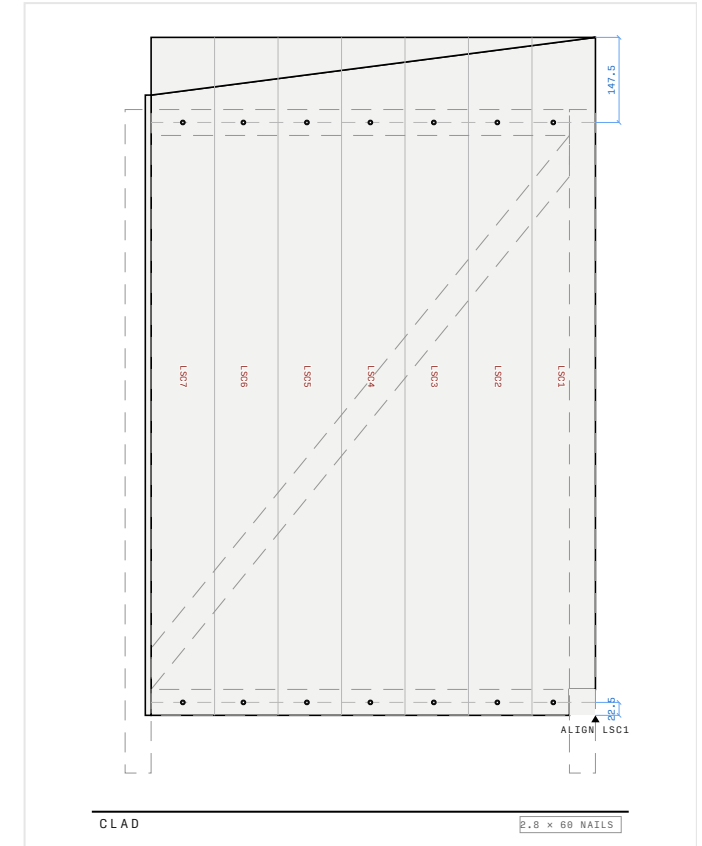


2 SCREW THE SIDE FRAME

• A-403.2

Match the frame diagonals before you drive a screw.

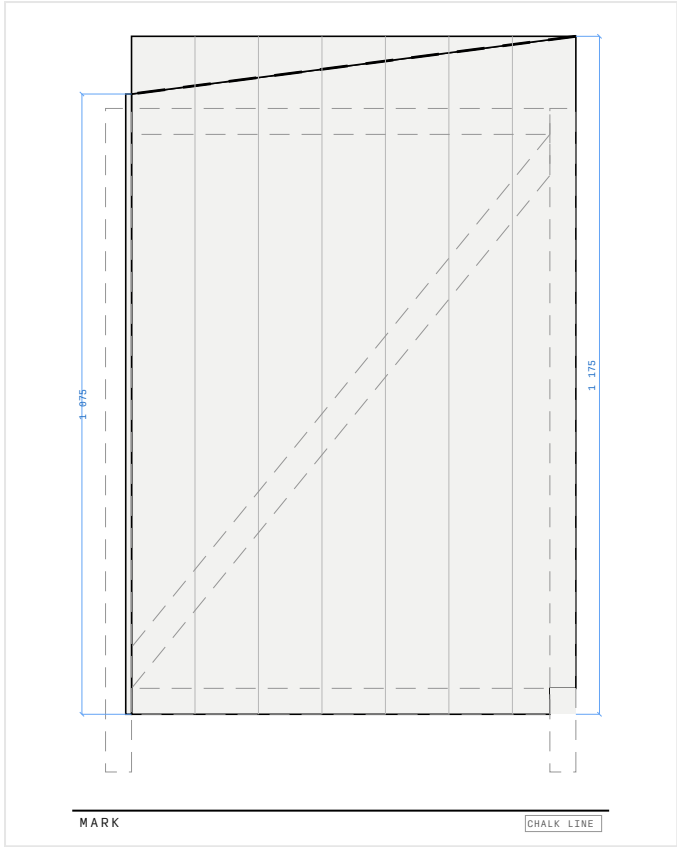
☐ DONE



3 FIX LSC1 TO LSC7, FRONT TO REAR

• A-403.3

☐ DONE

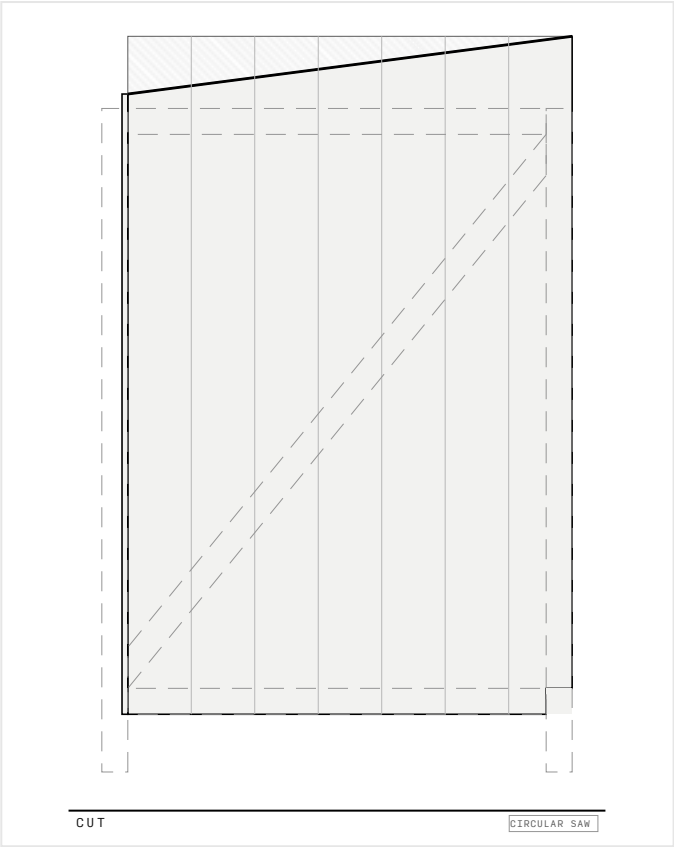


4 MARK THE SLOPE AT FRONT AND REAR

• A-403.4



DONE

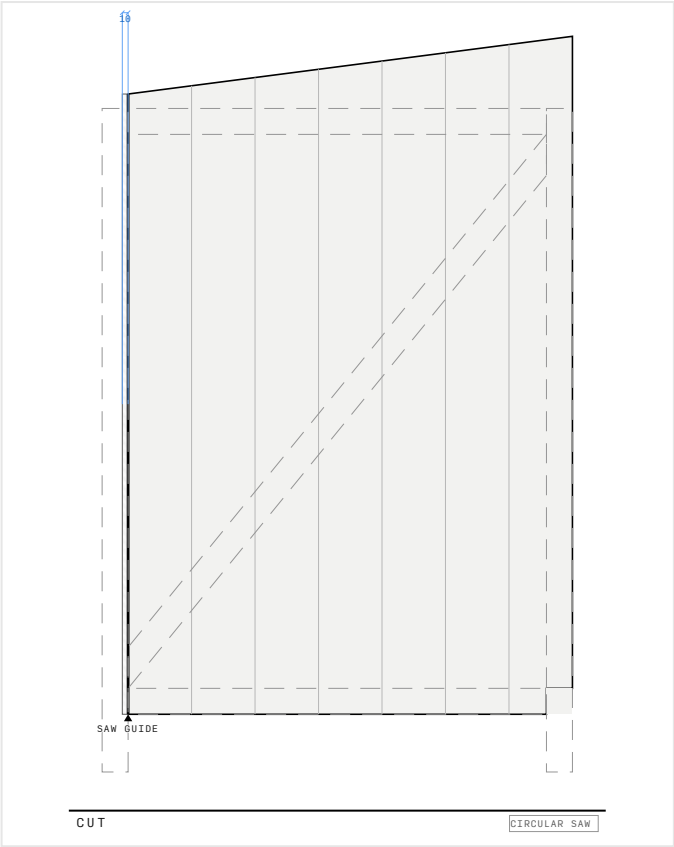


5 ONE GANG CUT ALONG THE MARK

• A-403.5



DONE

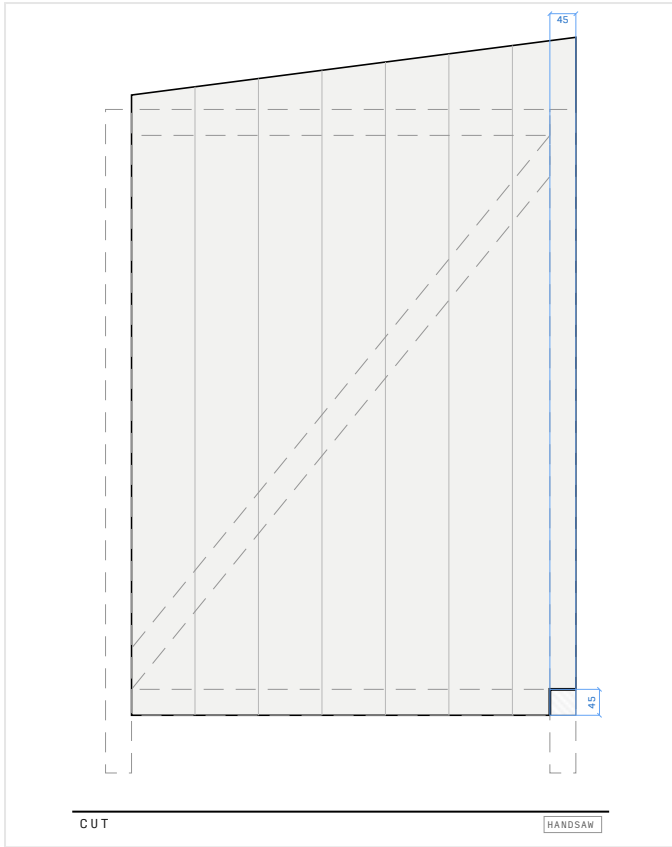


6 TRIM THE REAR OVERHANG

• A-403.6



DONE

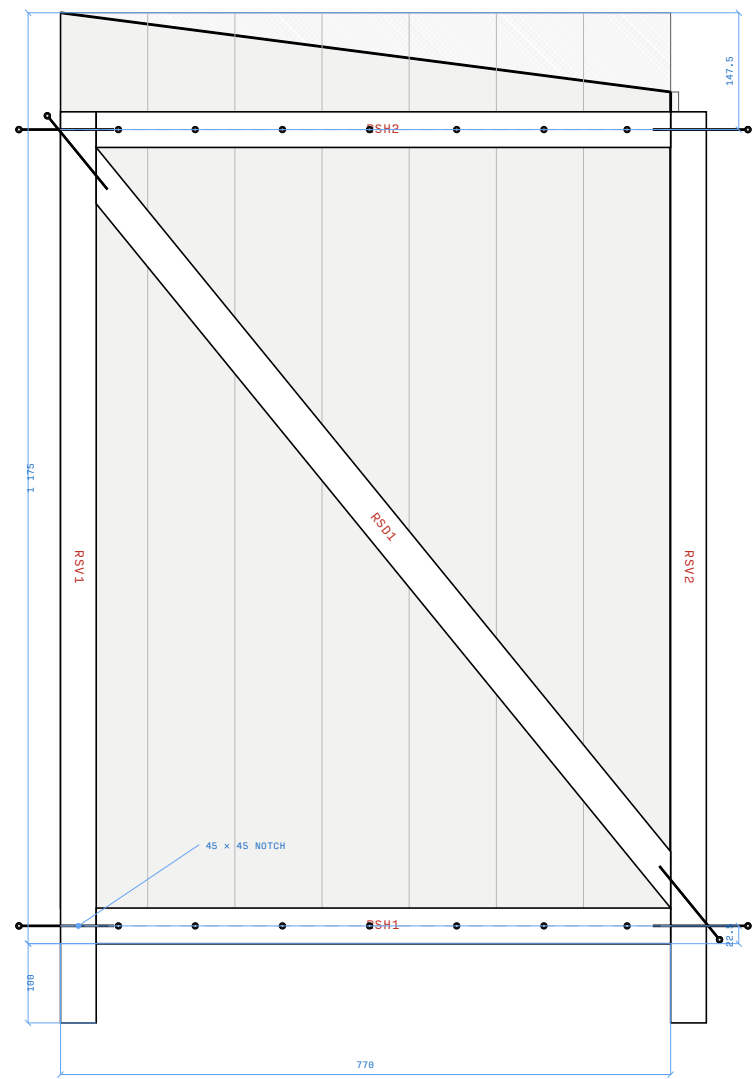


7 CUT THE BOTTOM-FRONT NOTCH

• A-403.7



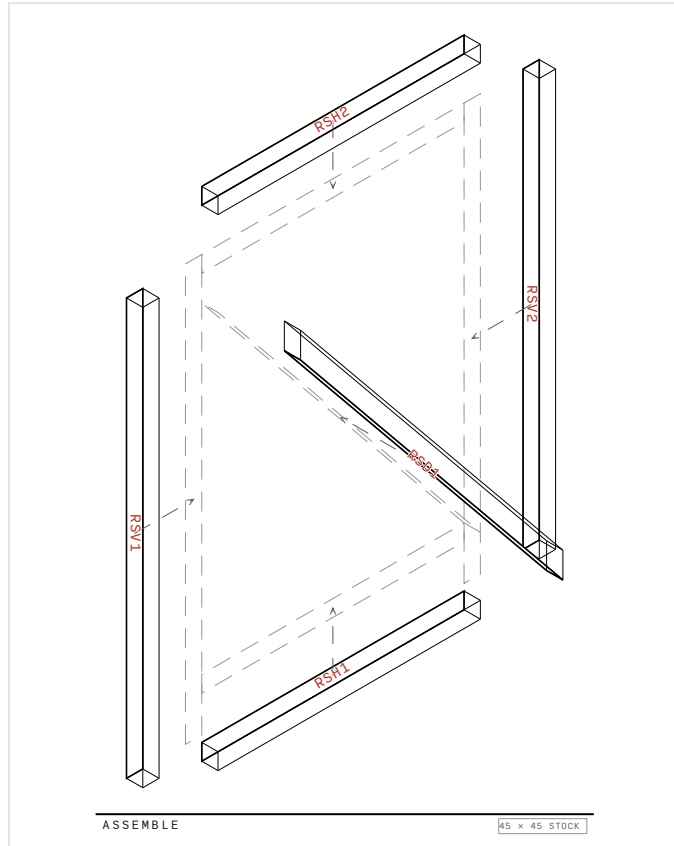
DONE



RIGHT SIDE UNIT • A-404 • RIGHT ELEVATION
front edge to the left • 100 fall over 770 • 45 x 45
bottom-front notch

HOLD

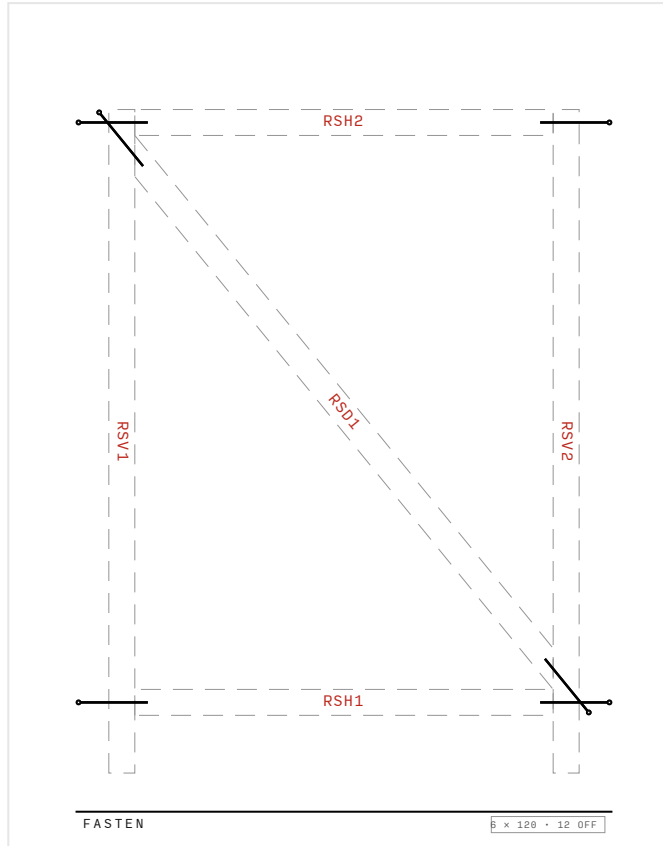
Do not pre-cut the roof reliefs. Hang and close the roof, then cut to the scribe.



1 BUILD THE SIDE FRAME

• A-404.1

☐ DONE

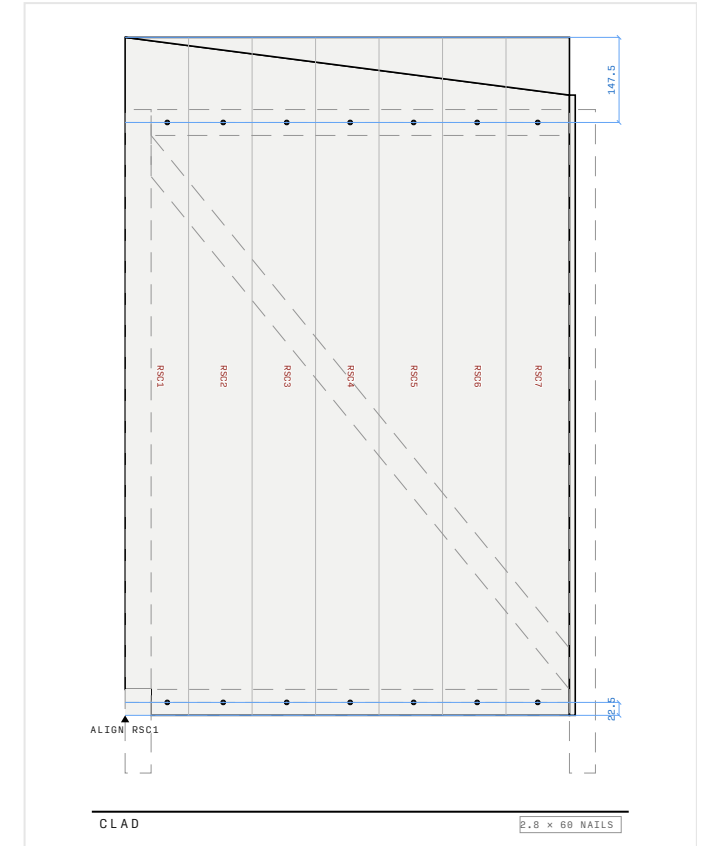


2 SCREW THE SIDE FRAME

• A-404.2

Match the frame diagonals before you drive a screw.

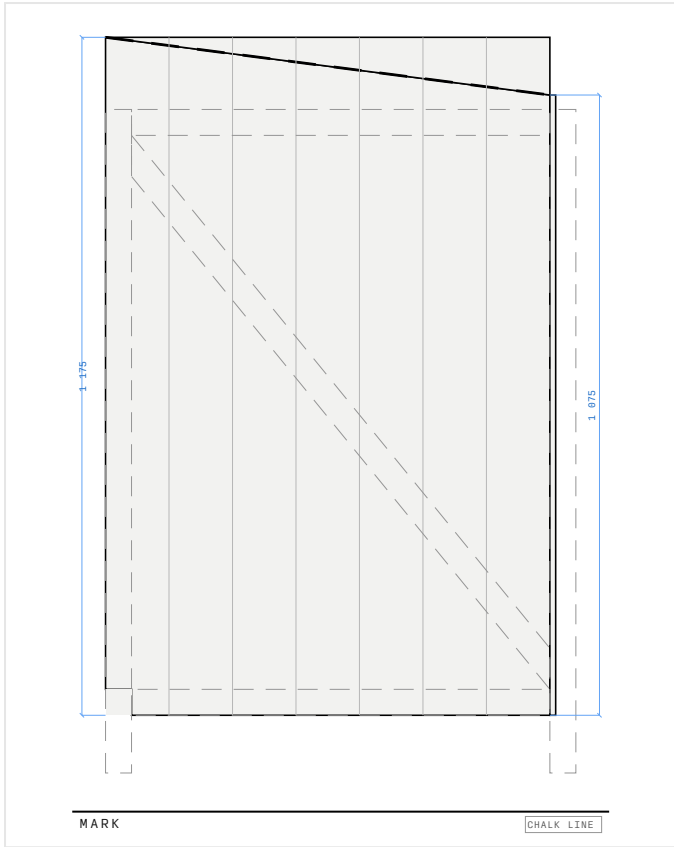
☐ DONE



3 FIX RSC1 TO RSC7, FRONT TO REAR

• A-404.3

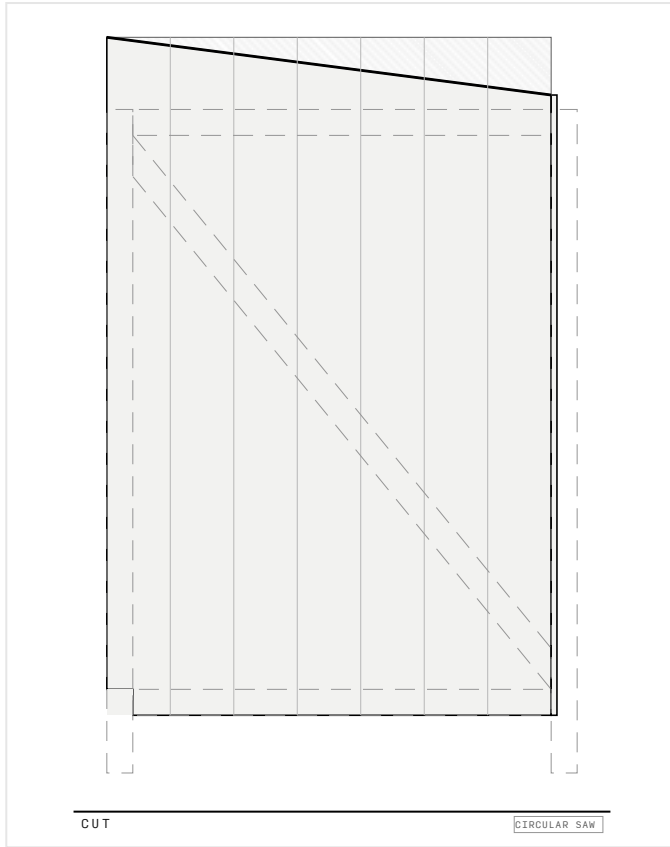
☐ DONE



4 MARK THE SLOPE AT FRONT AND REAR

• A-404.4

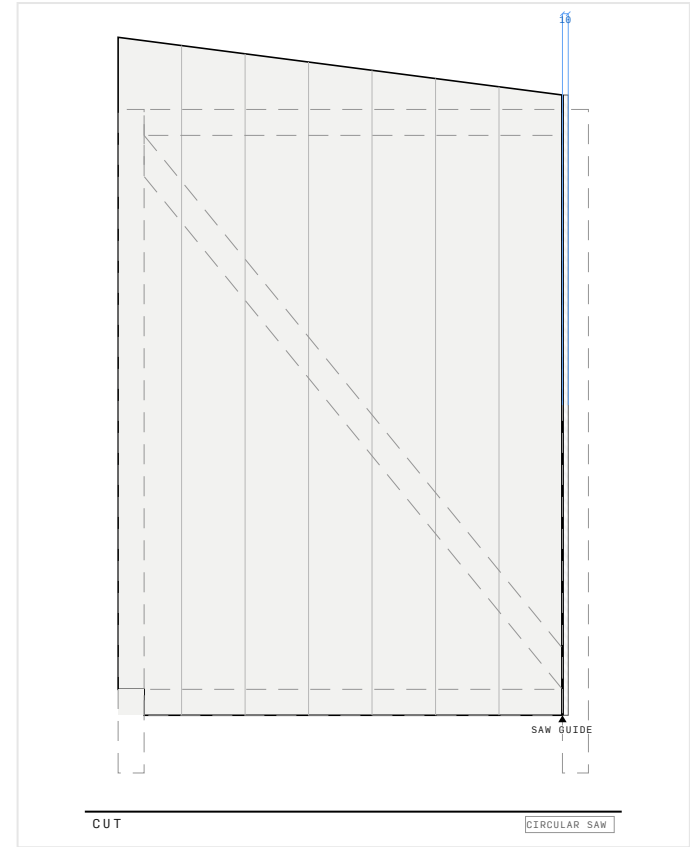
☐ DONE



5 ONE GANG CUT ALONG THE MARK

• A-404.5

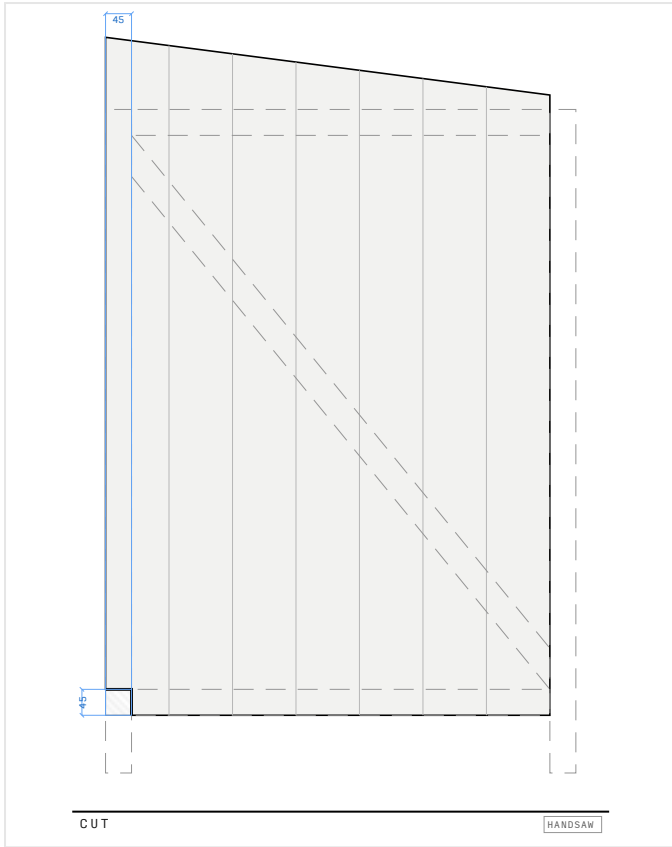
☐ DONE



6 TRIM THE REAR OVERHANG

• A-404.6

☐ DONE

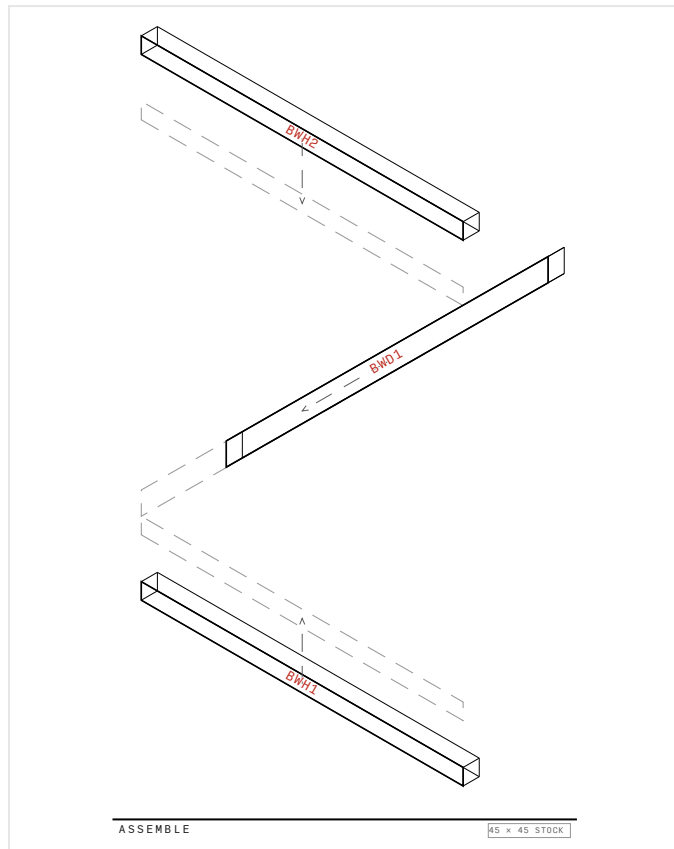


7 CUT THE BOTTOM-FRONT NOTCH

• A-404.7



DONE

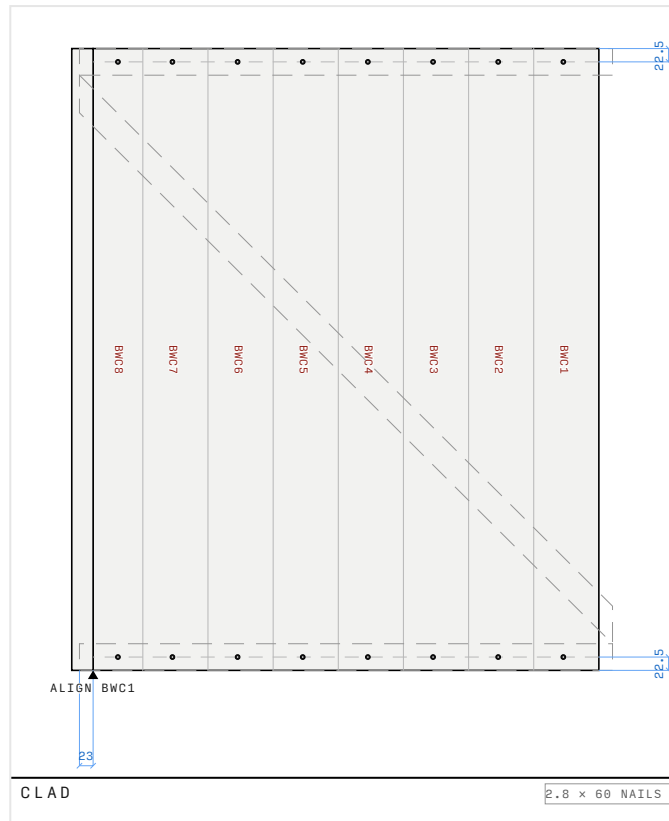


1 BUILD THE BARE BACK FRAME

• A-405.1

Hold it square while you clad it.
Nothing screws it until the posts do.

☐ DONE

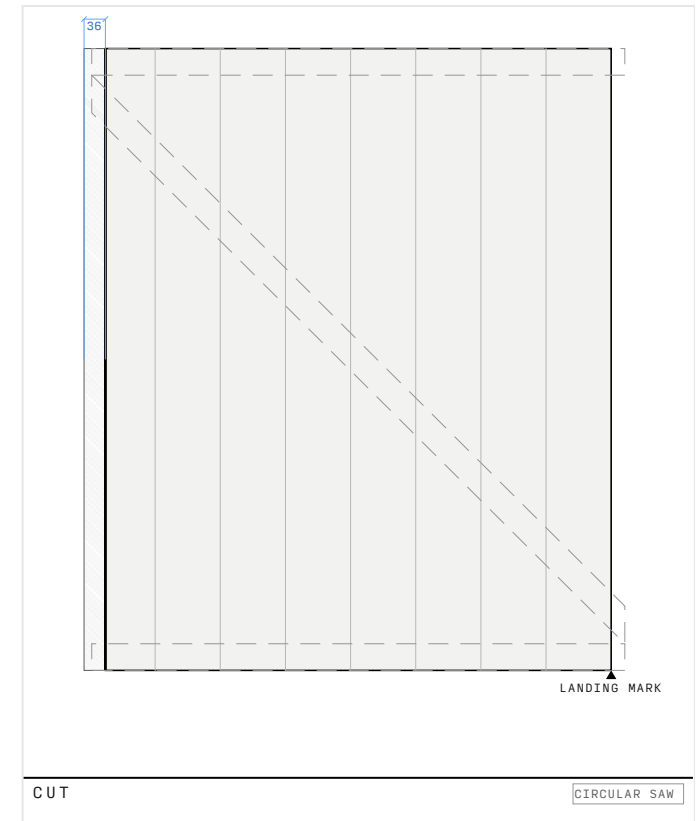


2 FIX BWC1 TO BWC8 TO THE LANDING MARKS

• A-405.2

Clad and trim the back flat. Fitted first, the side skins foul this overhang.

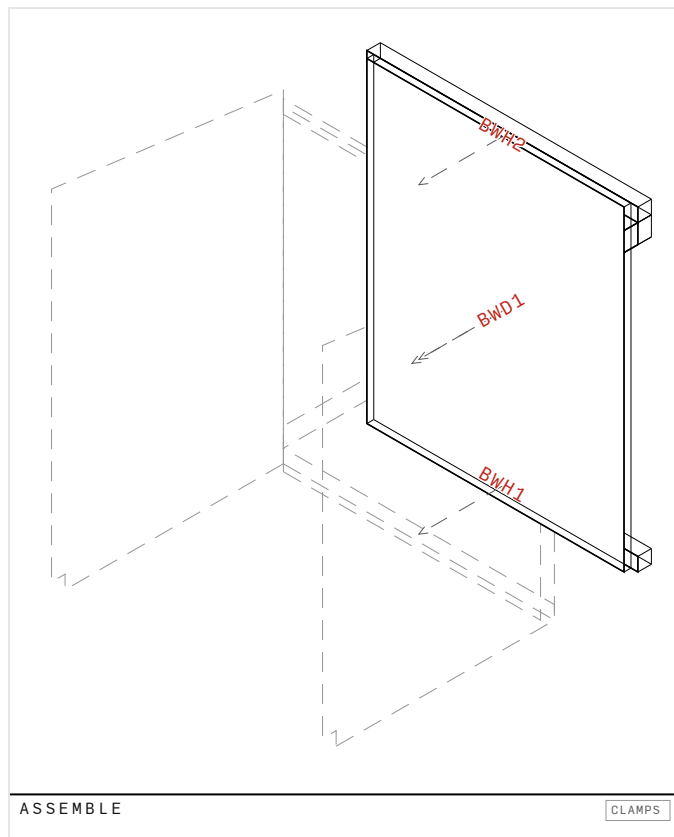
☐ DONE



3 TRIM THE RIGHT OVERHANG

• A-405.3

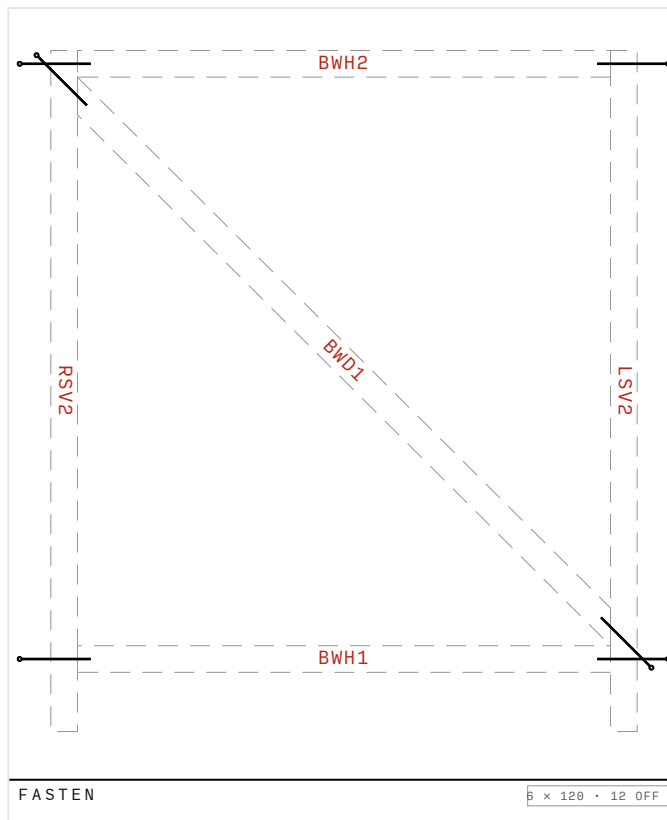
☐ DONE



4 SET IT BETWEEN THE TWO SIDE UNITS

• A-405.4

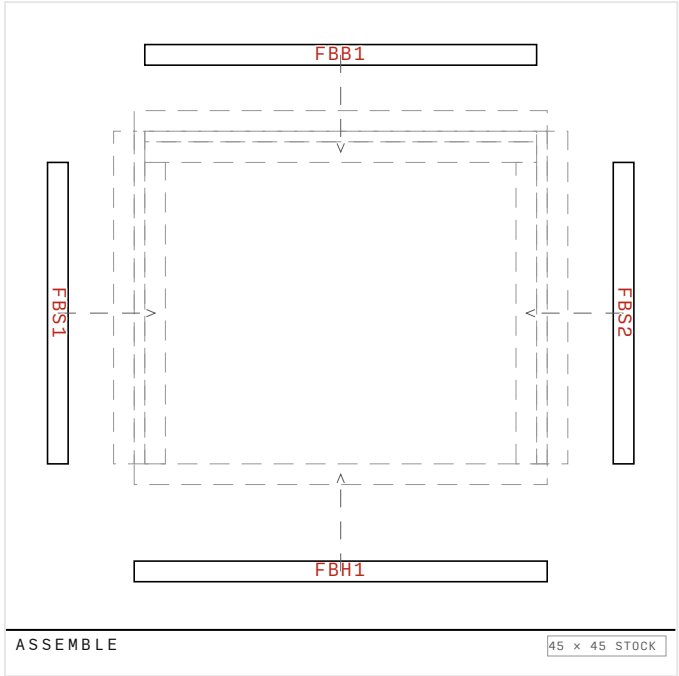
☐ DONE



5 SCREW THE REAR POSTS INTO BWH1 AND BWH2

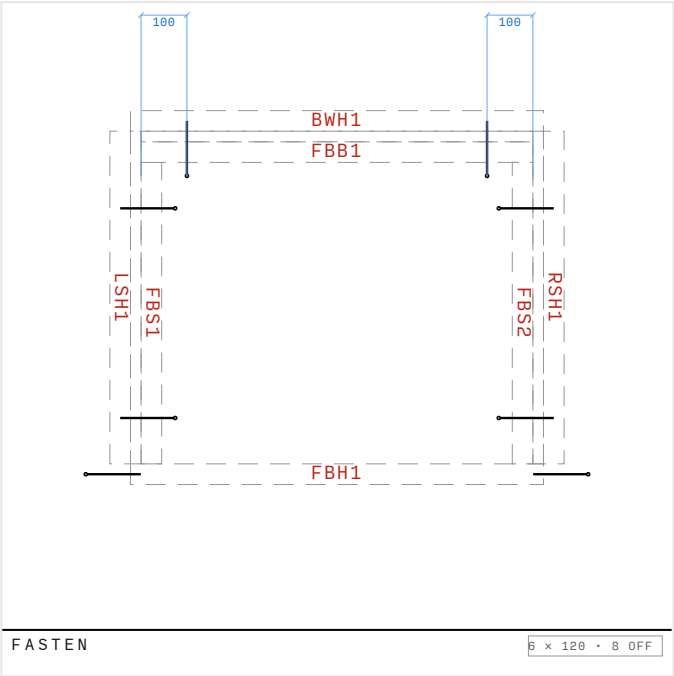
• A-405.5

☐ DONE



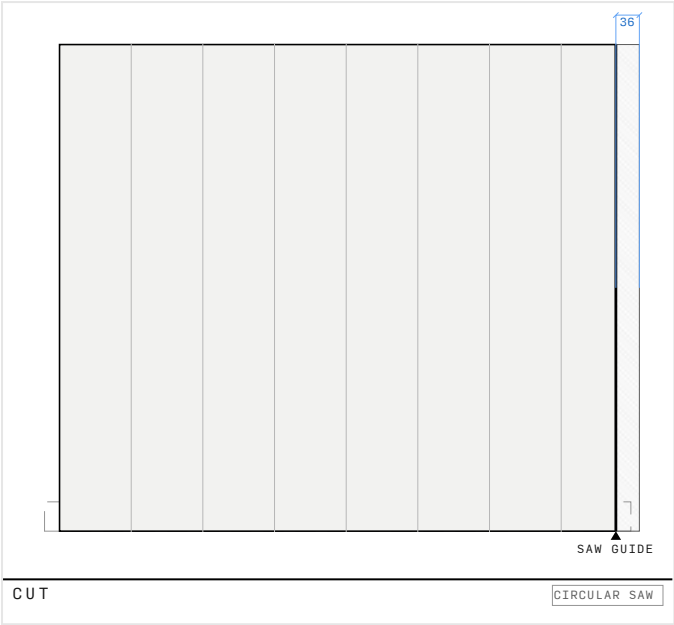
1 FIT FBB1, FBS1, FBS2 AND FBH1
• A-406.1

☐ DONE



2 SCREW ALL FOUR BEAMS TO THE SHELL
• A-406.2

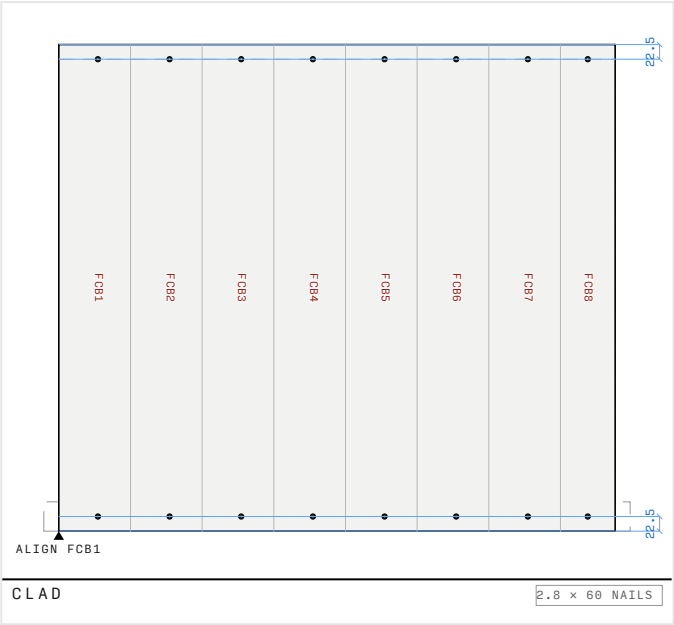
☐ DONE



1 TRIM THE RIGHT OVERHANG

- A-407.1

☐ DONE



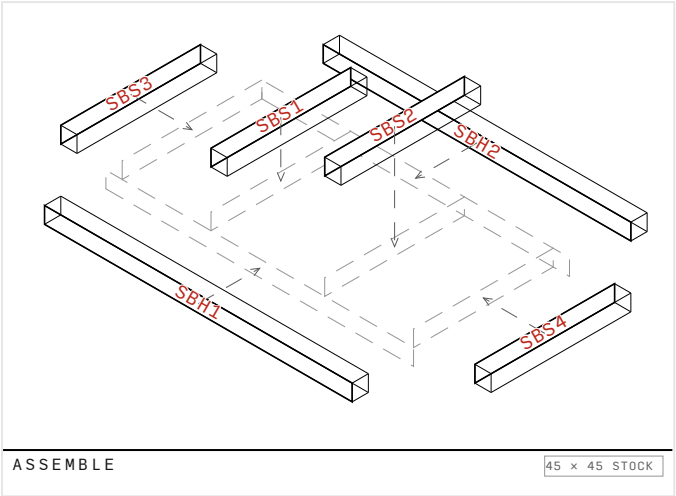
2 FIX FCB1 TO FCB8 TO THE BEARERS

- A-407.2

☐ DONE

HOLD

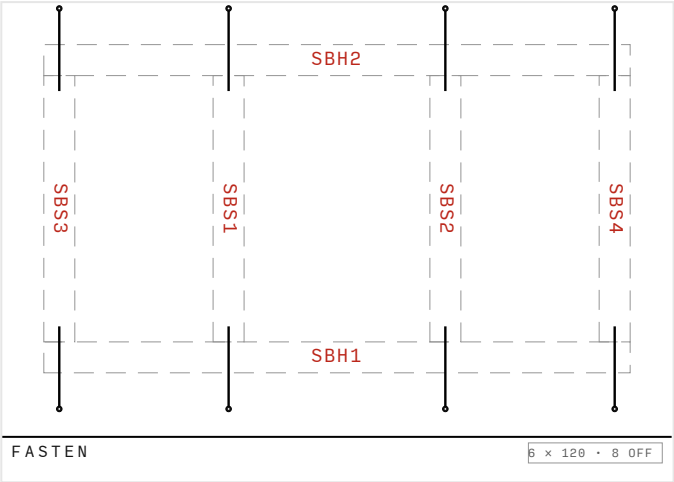
Seal every fresh cladding cut before the box goes in.



1 BUILD THE REMOVABLE SEAT BOX

• A-408.1

☐ DONE

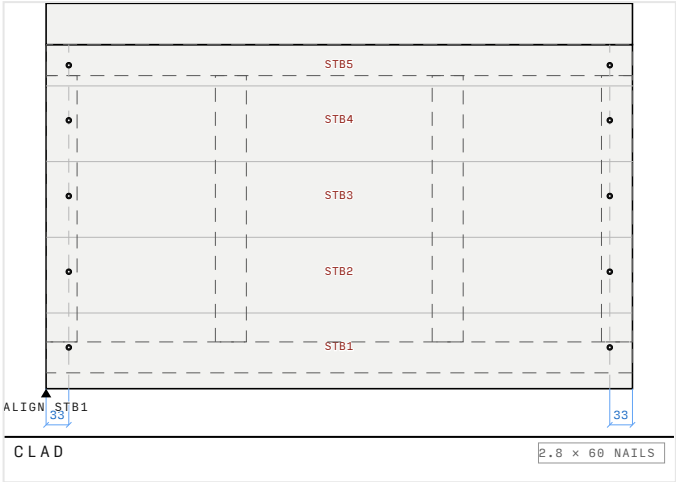


2 SCREW THE BOX FRAME

• A-408.2

Match the frame diagonals before you drive a screw.

☐ DONE



3 FIX STB1 TO STB5 ACROSS THE TOP

• A-408.3

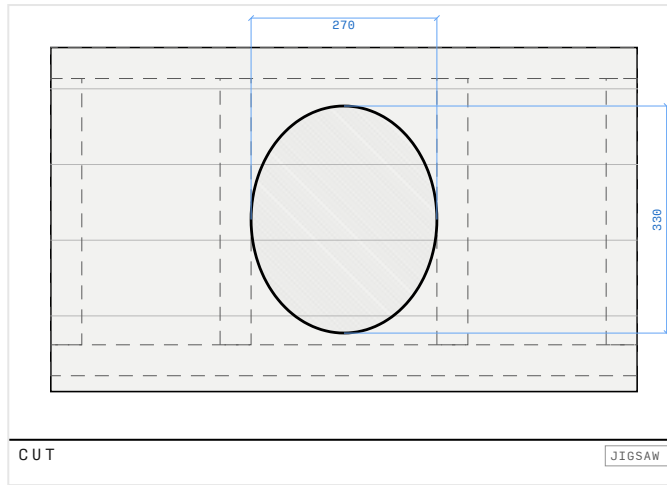
☐ DONE



4 TRIM THE FRONT OVERHANG

• A-408.4

☐ DONE



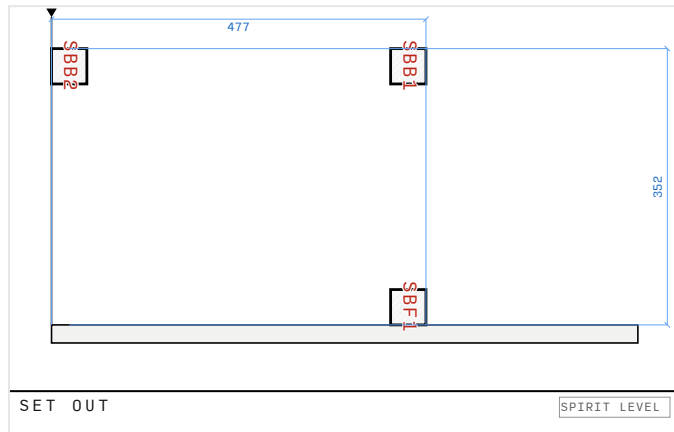
5 CUT THE SEAT OPENING

• A-408.5

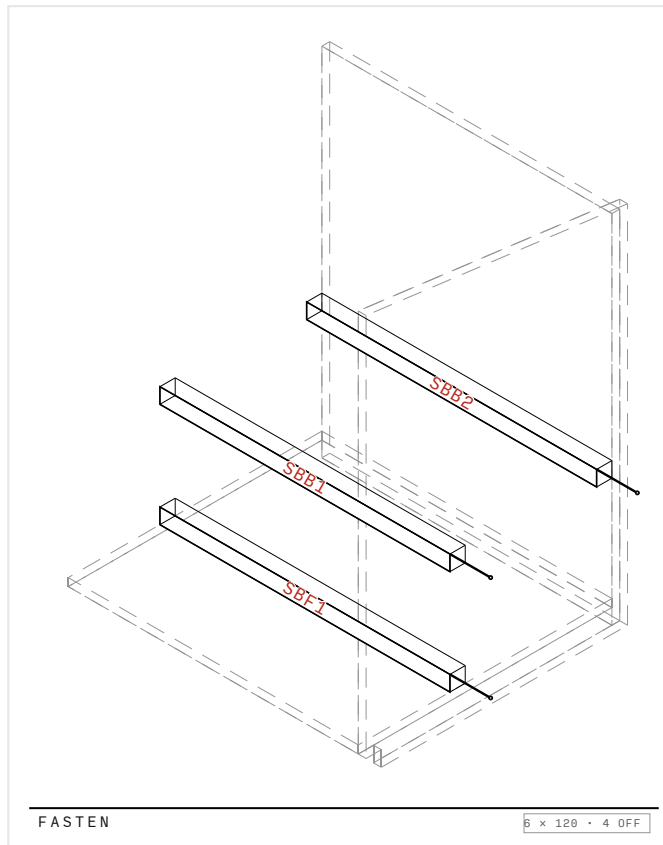
☐ DONE

HOLD

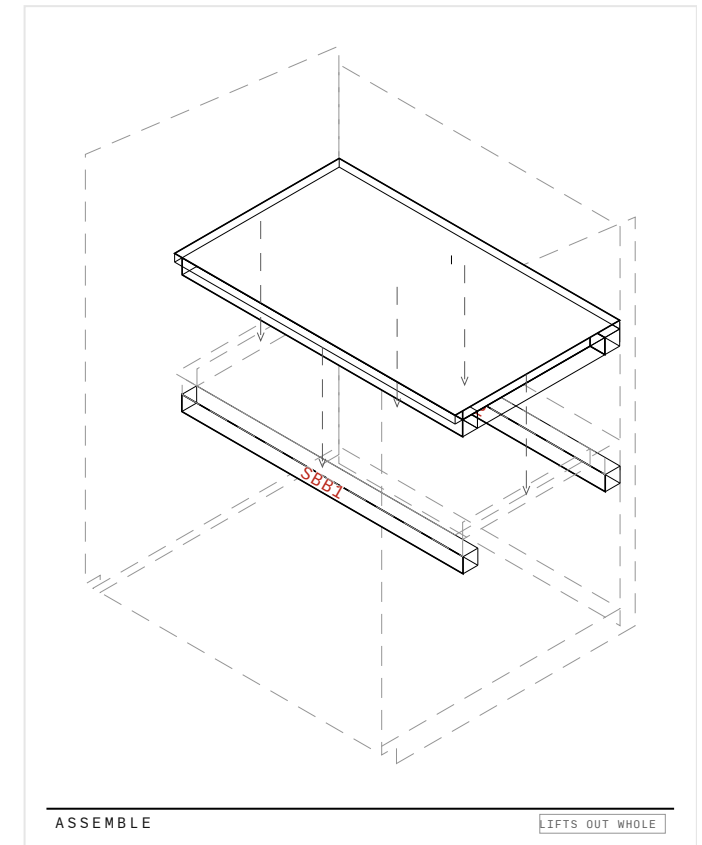
Leave both bearers screwed to the shell so the box lifts out as one assembly.

**1 SET BOTH BEAM TOPS ABOVE THE DECK**

• A-409.1

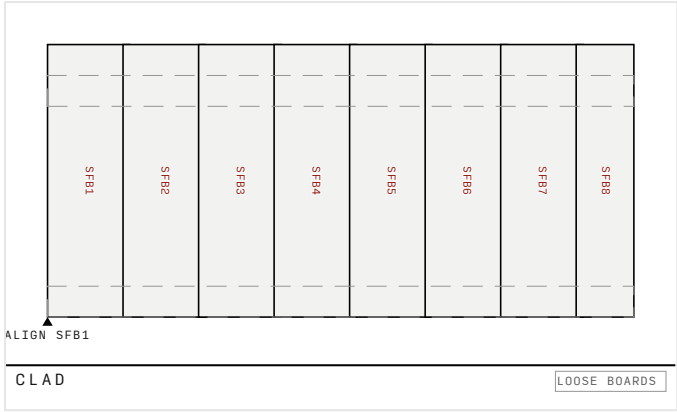
**2 DRIVE ONE SCREW FROM EACH SIDE**

• A-409.2

**3 LOWER THE BOX ONTO SBB1 AND SBB2**

• A-409.3

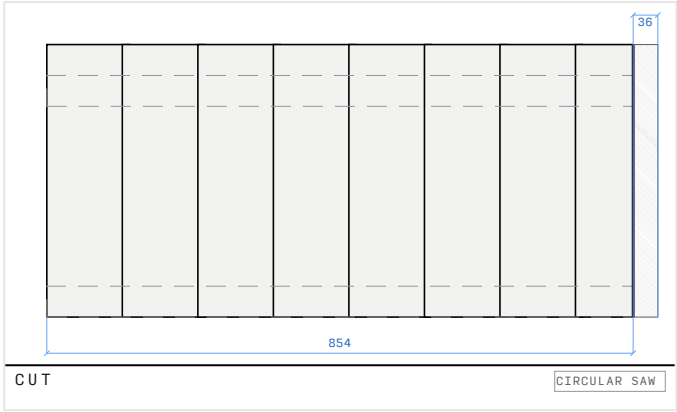
☐ DONE☐ DONE☐ DONE



1 LAY SFB1 TO SFB8 IN ORDER

• A-410.1

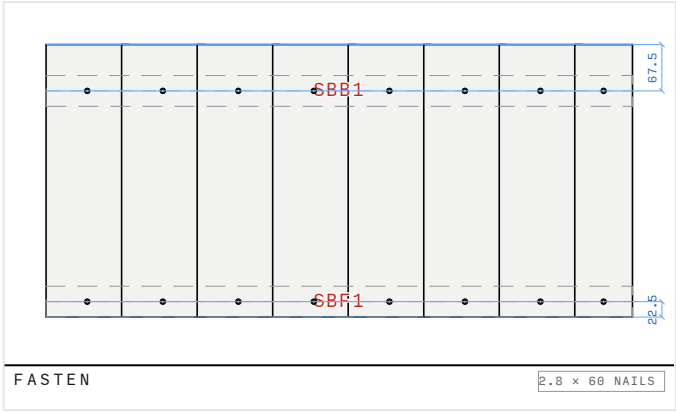
☐ DONE



2 TRIM THE FIELD TO THE OPENING WIDTH

• A-410.2

☐ DONE



3 SCREW EVERY BOARD TO SBF1 AND SBB1

• A-410.3

☐ DONE